

**ECSE-4040 DIGITAL ELECTRONICS
FINAL EXAM, SPRING 2026
RENSSELAER POLYTECHNIC INSTITUTE**

ASSIGNED: W, APRIL 15; DUE: W, APRIL 29

STATEMENT OF UNDERSTANDING

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"I understand that this Exam is officially administered under an Academic Honor Code: I hereby state that I have not used any other source material in completing this Exam other than my personal notes, the course notes, the course text, and/or SPICE software and documentation. In addition, I have not consulted or interacted, in any way or form, with anyone else, or any generative AI tool (such as ChatGPT), except, possibly, the Course Instructor, in completing this Exam."

Name: _____ (printed) Signature: _____

Student Number: _____ Date: _____

(Submit this signed page along with your work on LMS)

*Note, for all SPICE simulations using MOS transistors, use the provided **MOS2.txt** file on LMS.*

There are three problems on this exam. Use the given 130nm process parameters below for hand calculation for all problems where MOS transistors are used:

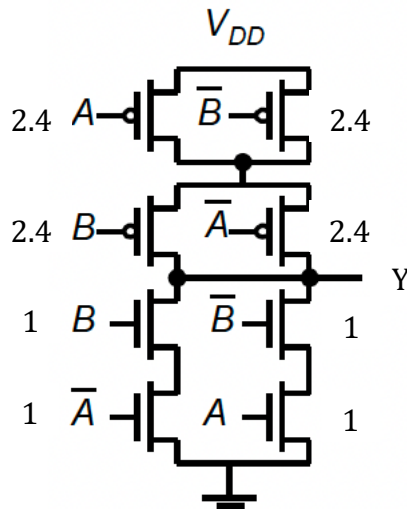
Below is a table of process parameters for a generic 0.13 μm technology.

0.13 μm Process technology parameters

Typical Technology Parameters	Symbol	NMOS	PMOS
Gate thickness	t_{ox} [Å]	22	22
Poly doping level	N_D [cm^{-3}]	3×10^{20}	—
Poly doping level	N_A [cm^{-3}]	—	3×10^{20}
Substrate doping level	N_A [cm^{-3}]	3×10^{17}	—
N-well doping level	N_D [cm^{-3}]	—	3×10^{17}
Number of surface state charges	N_{ss} [cm^{-2}]	6×10^{11}	6×10^{11}
Electron Mobility	μ_n [cm^2/Vs]	270	—
Hole Mobility	μ_p [cm^2/Vs]	—	70
Critical Field x Length	$E_C L$ [V]	0.6	2.4
Saturation velocity	v_{sat} [cm/s]	8×10^6	8×10^6
Gamma	γ [$\text{V}^{1/2}$]	0.2	0.2
Oxide capacitance	C_{ox} [F/ cm^2]	1.6×10^{-6}	1.6×10^{-6}
Threshold voltage	V_T [V]	0.4	− 0.4
2 x Fermi voltage	$2\phi_F$ [V]	−0.88	0.88
Channel length modulation	λ [V^{-1}]	0.2	0.2
Length	L [nm]	100	100

Problem 1 (30 pts. HAND, 10 pts. SPICE):

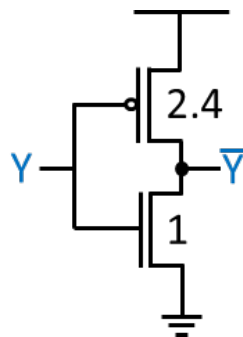
Consider the circuit below with the given 130nm process transistor parameters with the four given input transitions. R_{eff} values are given for a $W/L = 1$ scenario. Transistor W/L ratio is given in the schematic.



- $R_{Neff} = 12.5 \text{ k}\Omega$
- $R_{Peff} = 30 \text{ k}\Omega$
- $C_{self} = 1 \text{ fF}/\mu\text{m}$
- $C_g = 2 \text{ fF}/\mu\text{m}$
- $C_{int} = 0.1 \text{ fF}/\mu\text{m}$
- $L = 100 \text{ nm}$

1. $AB = 10 \rightarrow 00$
2. $AB = 01 \rightarrow 00$
3. $AB = 11 \rightarrow 10$
4. $AB = 11 \rightarrow 01$

- For all four input change scenarios, draw the equivalent Elmore Delay RC model for output node Y. Consider all transistors to have a constant resistance when on as given.
- Propagation delay calculation:
 - For all four input scenarios, use your Elmore Delay RC models to derive and solve formulas for the propagation delay for node Y.
 - Plot all 4 scenarios in LTSpice and state the simulated propagation delays. Compare these to your hand calculated results.
- Consider the prior circuit with an inverter added at node Y. For all for input patterns, what is the new propagation delay at node Y, and the total propagation delay to $\bar{Y}$?

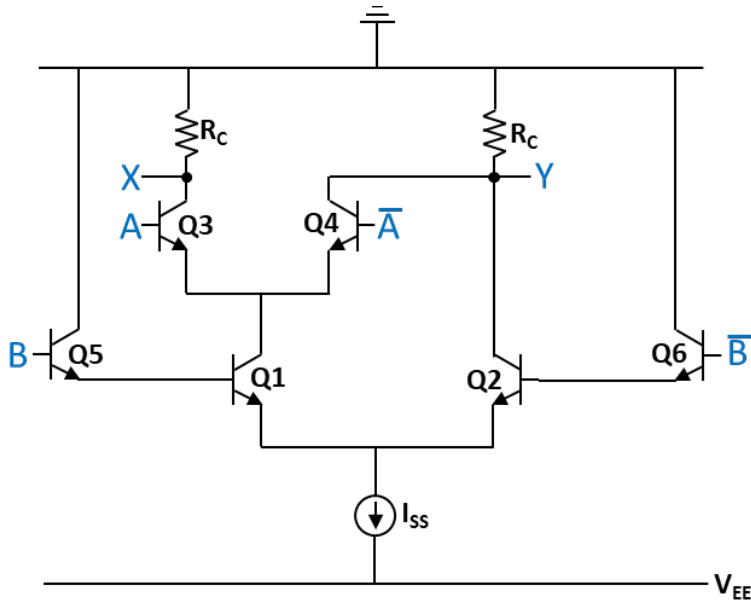


- A 10 mm long wire is added at node $\bar{Y}$. Considering this as a lumped, resistanceless capacitance, by hand calculate the delay for all four input change scenarios for nodes Y and $\bar{Y}$.
- If the widths of both transistors in the inverter were doubled, what happens to the propagation delays at nodes Y and $\bar{Y}$?
- What is the logic function at node Y? What purpose does this logic function perform?

Problem 2 (20 pts. HAND, 10 pts. SPICE):

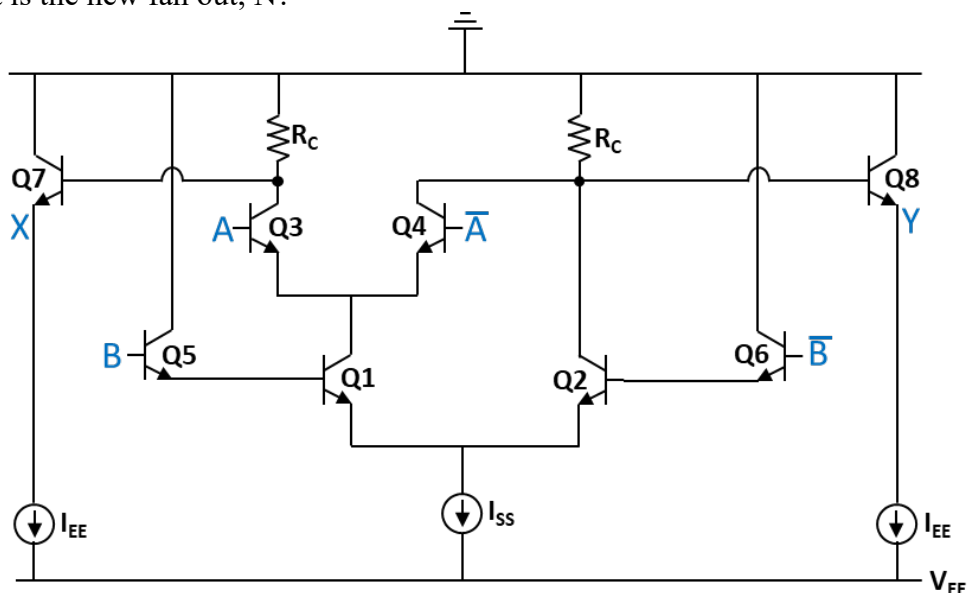
CML Logic & Fan out

Consider the circuit below:



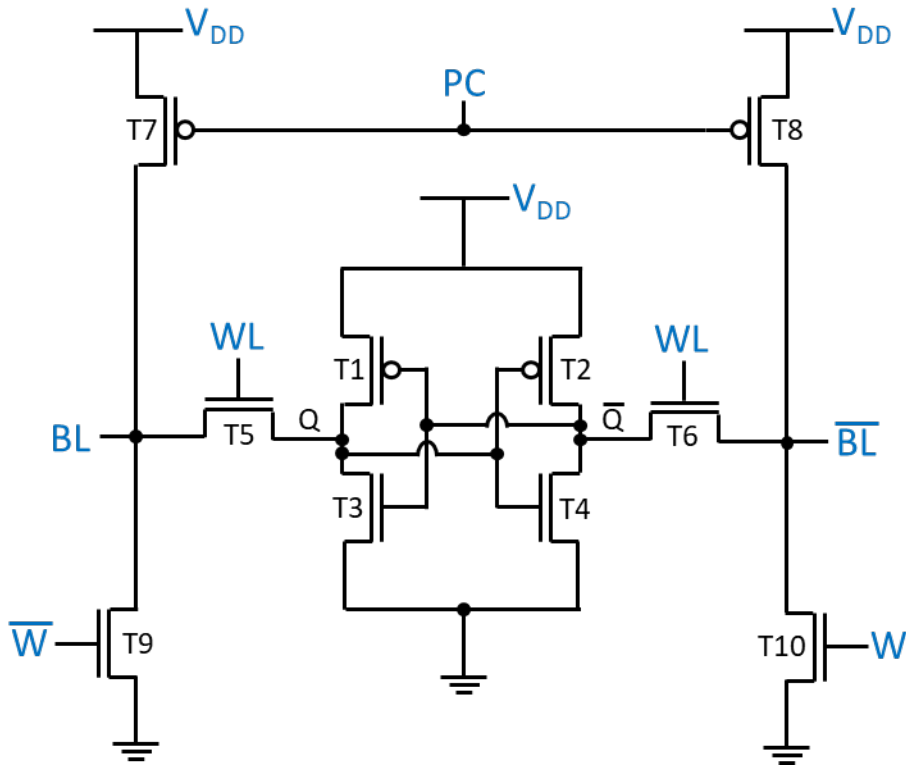
- $I_{SS} = 1 \text{ mA}$
- $V_{EE} = -5 \text{ V}$
- $R_C = 250 \Omega$
- $\beta_F = 100$
- $V_{BE}(FA) = 0.7 \text{ V}$
- $V_{BE}(\text{sat}) = 0.7 \text{ V}$
- $V_{CE}(\text{sat}) = 0.0 \text{ V}$

- Find V_{OH} , V_{OL} , V_{IH} , V_{IL} for the circuit above at output nodes X and Y considering a single input change.
- Plot the VTC with $\bar{A}$ and $\bar{B}$ fixed to V_{OL} , and B fixed to V_{OH} . Plot both outputs X and Y, with input A ranging from V_{EE} to 0, labelling V_{OH} , V_{OL} , V_{IH} , V_{IL} .
 - Plot both V_X and V_Y by hand.
 - Plot both V_X and V_Y with LTSpice. Compare with hand results.
- Find the noise margins, NM_H and NM_L for outputs X and Y.
- What are the logic functions at nodes X and Y?
- What is the purpose of transistors Q5 and Q6? Why can't inputs B and $\bar{B}$ be input directly to Q1 and Q2, respectively?
- Find the fan out, N, to input A, when $NM'_H = 0$.
 - Calculate by hand.
 - Verify fanout with LTSpice. Compare with hand results.
- Consider the following modified circuit with the additional Q7 and Q8 transistors, with $I_{EE} = 1 \text{ mA}$:
 - What are the new V_{OH} , V_{OL} values?
 - What is the new fan out, N?



Problem 3 (20 pts. HAND, 10 pts. SPICE):**SRAM Write**

Consider the following SRAM circuit with pre-charge and write circuitry with 130nm process parameters:



- $W_P/L = 2.4$ for all PMOS
- $W_N/L = 1$ for T3, T4, T5, T6
- $W_N/L = 10$ for T9, T10
- $L = 100\text{nm}$
- $V_{DD} = 1.2\text{V}$

- Identify the transistors that would pass current to perform a write of logic 1 to the SRAM cell.
- Assuming a successful write of logic 1, what states would these transistors be in?
- Assuming a **failed** write of logic 1 (the transistors cannot pull Q, $\bar{Q}$ below $V_{DD}/2$):
 - What states would the transistors responsible for a write of logic 1 be in?
 - Solve by hand the voltage obtained at node $\bar{Q}$. Assume $\bar{BL} = 0$.
 - Simulate this failed write in LTSpice.
 - In LTSpice, modify transistor widths so that a functional write is obtained. What transistor widths did you modify, and why?

Submit your work on LMS under **Final**

- Show your work.
 - Include images of your hand calculations with work and final results.
 - Include screenshots of LTSpice with all circuit schematics with all output plots.
- Due date: Wednesday, April 29th at 11:59pm.